ModelIt!

Teacher's Guide

Engineering a Superbug Killer

Optimizing Bacteriophage Therapy Against Antibiotic-Resistant Bacteria

Quick Reference

Grade Level	Grade 09L3
NGSS Standard	HS-LS1-2, HS-LS4-2
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-LS1-2, HS-LS4-2

Develop and use a model to illustrate the hierarchical organization of interacting systems that provide specific functions within multicellular organisms; construct an explanation based on evidence that the process of evolution primarily results from natural selection.

Three-Dimensional Learning

Science Practice	**Constructing Explanations and Designing Solutions** — Students construct evidence-based explanations of phage-bacteria co-evolution and design optimized therapeutic protocols using model-derived data.
Disciplinary Core Idea	**LS1.A Structure and Function / LS4.B Natural Selection** — Organisms have complex hierarchical structures that determine function; natural selection acts on heritable variation, driving bacterial resistance evolution under phage pressure.
Crosscutting Concept	**Cause and Effect / Stability and Change** — Students analyze the causal chain from phage infection to bacterial lysis to resistance evolution, and identify conditions that shift the system toward treatment success or failure.

CAST Testing Alignment

MULTIPLE CHOICE:

A patient's phage therapy is started 5 days after infection instead of 1 day. Based on your model, which outcome is most likely and why?

CONSTRUCTED RESPONSE:

Using your model, explain the 'arms race' between bacteriophages and bacteria. Why does bacterial resistance to phages sometimes make the bacteria more vulnerable to antibiotics? Reference at least three model components.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

This lesson introduces students to predictive modeling and optimization in the context of bacteriophage therapy — one of the most promising alternatives to antibiotics. Biotech skill focus: Predictive modeling and optimization. With antibiotic resistance killing over 1.27 million people annually and the pipeline for new antibiotics nearly dry, phage therapy represents a paradigm shift in infectious disease treatment that students must understand.

The Antibiotic Resistance Crisis

Antibiotic-resistant bacteria kill more people than HIV/AIDS or malaria. The WHO has declared antimicrobial resistance one of the top 10 global public health threats. Resistance arises through natural selection: when antibiotics are used, susceptible bacteria die while naturally resistant mutants survive and reproduce. Hospitals harbor 'superbugs' like MRSA (methicillin-resistant Staphylococcus aureus), CRE (carbapenem-resistant Enterobacteriaceae), and extensively drug-resistant tuberculosis that are resistant to nearly all available antibiotics. The pharmaceutical industry has largely abandoned antibiotic development because it's not profitable — a new antibiotic costs $1-2 billion to develop and is used sparingly (to prevent resistance), generating far less revenue than drugs taken daily for chronic conditions.

Bacteriophage Biology

Bacteriophages are viruses that exclusively infect bacteria — they cannot infect human cells because they require bacterial surface receptors for attachment. Phages follow a lytic cycle: (1) attachment to specific bacterial surface receptors, (2) injection of phage DNA into the bacterium, (3) hijacking of bacterial protein synthesis machinery to produce new phage components, (4) assembly of 50-200 new phage particles inside the cell, (5) lysis — the bacterial cell bursts, releasing new phages to infect more bacteria. This specificity is both a strength (precision targeting) and a challenge (each phage only kills specific bacterial strains). Phage cocktails — mixtures of multiple phage types — address this limitation.

Phage-Bacteria Co-Evolution: The Red Queen Hypothesis

Bacteria and phages have been co-evolving for 3.5 billion years in the longest-running arms race on Earth. Bacteria evolve phage resistance primarily by modifying surface receptors (so phages can't attach), producing restriction enzymes (that cut foreign DNA), or using CRISPR-Cas systems (bacterial immune memory). But resistance often comes at a cost: the surface receptors that phages use for attachment are frequently the same receptors bacteria use for virulence or antibiotic resistance. When bacteria modify these receptors to escape phages, they may lose their antibiotic resistance — a phenomenon called 'phage-antibiotic synergy' or 'evolutionary trade-offs.' This creates a therapeutic strategy: use phages to force bacteria into an evolutionary corner where they become vulnerable to antibiotics again.

Clinical Phage Therapy: From Compassionate Use to Precision Medicine

Phage therapy has been used clinically in the Republic of Georgia for over 100 years, but Western medicine largely abandoned it when antibiotics were discovered. Now, with antibiotics failing, phage therapy is experiencing a renaissance. The FDA has approved compassionate-use phage therapy for patients with untreatable infections. Notable cases include Tom Patterson (2016), who survived a multi-drug-resistant Acinetobacter infection after personalized phage therapy when all antibiotics failed. Modern phage therapy uses genomic sequencing to match phages to patient-specific bacterial strains, synthetic biology to engineer phages with expanded host ranges, and computational models to optimize dosing and treatment timing. The challenge is that each treatment is essentially a custom-designed biological product — very different from mass-produced antibiotic pills.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify nine components spanning phage biology, bacterial evolution, immune response, and treatment parameters — including three externally controllable variables (Phage Specificity, Treatment Timing, Dosage Optimization) and six internal dynamic responses.

E **Establish Relationships**

Students determine that Dosage Optimization establishes the initial Phage Population, which interacts with Target Bacteria Population through predator-prey dynamics while Bacterial Resistance Rate and Immune System Response work against the therapy — and that Phage Specificity controls the trade-off between targeting precision and Healthy Microbiome Impact.

V **Visualize & Model**

Students build a computational model showing the three-way race between phage replication, bacterial resistance evolution, and immune clearance, with treatment outcome determined by which process dominates.

E **Evaluate Outcomes**

Students run standard, delayed-treatment, and optimized-cocktail scenarios to identify the parameter combinations that maximize bacterial clearance while minimizing resistance emergence and microbiome damage.

R **Revise & Extend**

Students add Phage-Antibiotic Synergy, Biofilm Penetration, or Phage Engineering Level to explore advanced therapeutic strategies.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Antibiotic-resistant bacteria killed 1.27 million people last year — more than HIV and malaria combined. The drugs are failing. But there's an ancient weapon we forgot about for 80 years: viruses that eat bacteria."*

Watch For

Show the WHO antibiotic resistance infographic. Let students react to the death toll before introducing phages.

Slide 2: Learning Objectives

3 min

Say: *"Today you're going to engineer a treatment protocol using something that's been killing bacteria for 3.5 billion years — bacteriophages. And you're going to optimize it using computational modeling."*

Watch For

Pre-teach bacteriophage and antibiotic resistance. Show a 30-second phage lysis animation to establish the biological mechanism.

Slide 3: Big Question

3 min

Say: *"Every antibiotic we've ever invented, bacteria have evolved resistance to. Every single one. So why would phage therapy be any different — or would it?"*

Watch For

Quick-write: Students predict whether bacteria can evolve resistance to phages and what might be different about viral versus chemical treatment. Share predictions.

Slide 4: LEVER Framework

2 min

Say: *"We're going to optimize a living treatment — one that reproduces, evolves, and interacts with the patient's immune system in real time. This is nothing like designing a pill."*

Watch For

Preview LEVER steps. Emphasize that phage therapy involves optimizing a dynamic biological system, not just finding the right dose.

Slide 5: Activity 1: Components

8 min

Say: *"Three of these components are under the doctor's control. Six are determined by biology and evolution. Which is which — and why does that distinction matter?"*

Watch For

Guide sorting of external versus internal components. Discuss why Phage Specificity is controllable through engineering but Bacterial Resistance Rate is not.

Slide 6: Activity 2: Connections

8 min

Say: *"This model has something none of our previous models had: the treatment reproduces inside the disease. Trace what happens when phages meet bacteria."*

Watch For

Students map the self-amplifying feedback loop. Identify the three-way race between phage replication, bacterial resistance, and immune clearance.

Slide 7: Activity 3: Simulation

10 min

Say: *"You're the biotech engineer. Find the combination of specificity, timing, and dosage that clears the infection. But watch what evolution does in real time."*

Watch For

Students predict outcomes, then run all three scenarios. Track bacterial clearance rate, resistance emergence, and microbiome impact as key metrics.

Slide 8: Discoveries

5 min

Say: *"Here's the twist: when bacteria evolve resistance to phages, they sometimes lose their antibiotic resistance. The phage forces them into an evolutionary trap. This is phage-antibiotic synergy — and your model just predicted it."*

Watch For

Connect model findings to the real evolutionary trade-off in phage therapy. Discuss how this changes the treatment strategy.

Slide 9: STEM Challenge

12 min

Say: *"A patient walks into your phage therapy clinic with an infection that no antibiotic can touch. Design their treatment protocol."*

Watch For

Groups design personalized protocols including phage selection, cocktail composition, timing, dosage, and resistance monitoring. Present and defend.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Phage Specificity, Treatment Timing, Dosage Optimization
- **INTERNAL:** Phage Population, Target Bacteria Population, Bacterial Resistance Rate, Immune System Response, Phage Replication Rate, Healthy Microbiome Impact

Relationships

- **Dosage Optimization to Phage Population:** POSITIVE (+) — Higher initial phage dose establishes a larger starting Phage Population, increasing the probability that phages encounter and infect target bacteria before the immune system clears them.
- **Phage Population to Target Bacteria Population:** NEGATIVE (-) — More phages infecting and lysing target bacteria directly reduces the Target Bacteria Population through predator-prey dynamics.
- **Target Bacteria Population to Phage Replication Rate:** POSITIVE (+) — More abundant target bacteria provide more hosts for phage replication, increasing Phage Replication Rate — this is the self-amplifying nature of phage therapy.
- **Phage Specificity to Healthy Microbiome Impact:** NEGATIVE (-) — Higher Phage Specificity means the phages more precisely target only the pathogenic bacteria, reducing off-target damage to beneficial gut microbiome species.
- **Target Bacteria Population to Bacterial Resistance Rate:** POSITIVE (+) — Larger bacterial populations contain more genetic diversity through random mutation, increasing the probability that resistance-conferring mutations exist and are selected for under phage pressure.

Key Discoveries

- Phage therapy is self-amplifying — phage populations grow exponentially when target bacteria are abundant, then decline naturally when the infection clears, making it inherently self-limiting
- Bacterial resistance to phages evolves rapidly but often comes with a fitness cost — resistant bacteria may lose virulence factors or become re-sensitized to antibiotics
- The immune system is both ally and obstacle — it helps clear the infection but may also neutralize therapeutic phages before they complete their work
- Treatment timing is critical because early intervention targets smaller, more genetically homogeneous populations that are less likely to contain pre-existing resistant mutants

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Phages are dangerous because they're viruses"	Bacteriophages CANNOT infect human cells — period. They require specific bacterial surface receptors that human cells do not have. Phages have been part of the human microbiome since birth; you carry approximately 10^12 phages in your gut right now. They are the most abundant biological entities on Earth and have never caused a human infection. The word 'virus' triggers fear, but phages are fundamentally different from human-infecting viruses.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Bacteria can become resistant to everything eventually"	While bacteria can evolve resistance to any single selective pressure, resistance always comes at a cost. The model shows that phage resistance often requires modifying surface receptors that serve other essential functions. Phage cocktails (multiple phages targeting different receptors) make it exponentially harder for bacteria to evolve resistance to ALL phages simultaneously — the same principle behind HIV combination therapy.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"We should just develop new antibiotics instead"	New antibiotic development has nearly stopped because it's economically unviable — $1-2 billion to develop a drug that's used sparingly and becomes obsolete when resistance emerges. Only 2 new antibiotic classes have been discovered in the last 50 years. Phage therapy offers a fundamentally different approach: instead of chemical warfare against bacteria, it uses a co-evolved biological weapon that adapts alongside its target.